

PRODUCT REQUIREMENTS DOCUMENT (PRD)

1. Product Overview

Product name: SarcoScan

One-line pitch: An on-premise AI screening tool that combines a routine knee AP X-ray with a 10-second digital handgrip test to flag sarcopenia and osteoporosis risk, with no DEXA or CT needed.

Platforms: Web dashboard (clinic and admin use) and mobile app (grip capture, quick screening, patient follow-up).

2. Problem Statement

Sarcopenia is under-diagnosed in elderly patients because confirmation needs DEXA or CT, which Tier-2 and Tier-3 hospitals often lack. Knee X-rays and handgrip tests are already done routinely, but nobody combines them to assess muscle health. Patients are diagnosed late, after falls or fragility fractures.

3. Goals

1. Identify patients at risk of sarcopenia using only routine data.
2. Perform better than handgrip strength alone (primary success metric).
3. Give an opportunistic osteoporosis risk estimate from the same X-ray.
4. Generate a standardized, risk-stratified clinical report.
5. Run on affordable on-premise hardware with no GPU.
6. Fit into an outpatient workflow in under 5 minutes per patient.

4. Non-Goals

- It does not replace DEXA for confirmed diagnosis. It is a screening and referral tool.
- It does not diagnose osteoarthritis (the KL grade is an informational bonus only).
- It does not offer treatment prescriptions.

5. Target Users and Personas

Persona	Description	Main needs
Orthopedic or geriatric doctor	Reviews results and decides on referral	Clear risk tier, explainable overlay, printable report
Radiology technician / nurse	Captures grip and uploads X-ray	Fast, simple, error-free data entry

Hospital admin	Manages users, devices, audit	Access control, usage stats, data security
Patient or caregiver (mobile, read-only)	Views own report and trend	Simple language, next steps

6. Key Features

6.1 Must-have (MVP)

1. **Secure login and role-based access** (doctor, technician, admin).
2. **Patient registration** (name, age, sex, height, weight, contact, optional ABHA ID).
3. **Knee X-ray upload** (DICOM, JPG, PNG) from web, or auto-ingest from Orthanc.
4. **Image quality check** (confirms knee AP view and that both soft-tissue edges are visible; warns if cropped).
5. **AI segmentation** of femur, tibia, soft-tissue envelope, with overlay.
6. **Muscle proxy measurement** (soft-to-bone ratios at thigh and calf levels).
7. **Handgrip entry: technician left aur right hand ke 3 trials manually enter karta hai, system best value choose karta hai aur AWGS cutoff se compare karta hai.**
8. **Clinical inputs form** (SARC-F, 5-chair-stand time, calf circumference, optional).
9. **Fusion model** giving sarcopenia probability and AWGS-style stage: No / Possible / Probable / Severe.
10. **Osteoporosis risk** as a calibrated probability (T-score at or below -2.5 equivalent) from the proximal tibia texture.
11. **Explainability** (Grad-CAM heatmap, measured values vs AWGS cutoffs).
12. **Standardized PDF report** with risk tier and suggested actions.
13. **Patient history and trend graph** across visits.
14. **Audit log** of every access and action.

6.2 Should-have

- KL osteoarthritis grade (bonus output).
- Orthanc PACS integration demo (DICOM C-STORE, DICOMweb).
- FHIR DiagnosticReport export.
- Dual-knee X-ray auto-crop (left and right analysed separately).
- Multi-language UI (English, Hindi).
- Offline mode on mobile with sync when online.

6.3 Nice-to-have

- ABHA/ABDM linking.
- Referral letter generator.
- Hospital-level analytics dashboard (screened count, risk distribution).
- WhatsApp or SMS report delivery.

7. Functional Requirements (detail)

ID	Requirement
FR-1	System shall accept DICOM and standard image formats up to 50 MB.
FR-2	System shall reject non-knee-AP images and show a reason.
FR-3	System shall flag images where the soft-tissue edge is cropped.
FR-4	System shall compute muscle features within 5 seconds per study on CPU.
FR-5	System shall record grip strength for left and right hand, 3 trials each, store best value.
FR-6	Grip cutoffs applied per AWGS 2019: below 28 kg (male), below 18 kg (female).
FR-7	System shall output sarcopenia probability, stage and confidence.
FR-8	System shall output osteoporosis risk probability and tier (Low / Moderate / High).
FR-9	Every report shall include input values, cutoffs, overlay, and disclaimer.
FR-10	Doctors may override or annotate the AI result; both are stored.
FR-11	Records are retained for longitudinal tracking and are exportable.

8. Non-Functional Requirements

- **Performance:** under 5 s inference on an i5 CPU with 16 GB RAM, no GPU.
- **Security:** encryption at rest and in transit, RBAC, audit logs, DPDP Act 2023 alignment.
- **Availability:** works on a local hospital network without internet.
- **Usability:** a trained technician completes a screening in under 5 minutes.
- **Accessibility:** large fonts, high contrast, usable on tablets.
- **Scalability:** supports 200 screenings per day per site.

9. Success Metrics

Metric	Target (pilot)
AUC of fusion model vs grip alone	Fusion higher, statistically significant (DeLong test)
Sensitivity at screening threshold	85% or higher
Osteoporosis AUC vs DXA labels	0.80 or higher
Time per patient	Under 5 minutes

Report generation time	Under 10 seconds
Clinician satisfaction	4 out of 5 or higher

Note: state these as targets to validate, not claimed results.

10. Risks and Mitigations

Risk	Mitigation
X-ray is a weak muscle signal; fat looks like muscle	Add BMI and grip strength to model; be transparent in the report
Limited labeled data with paired DXA	Weak labels from OAI thigh MRI, then fine-tune on partner-hospital data
Cropped X-rays	Quality check step with warning
Clinician mistrust of AI	Grad-CAM, shown cutoffs, override option
Data privacy	On-prem deployment, encryption, anonymized training data

11. Release Plan

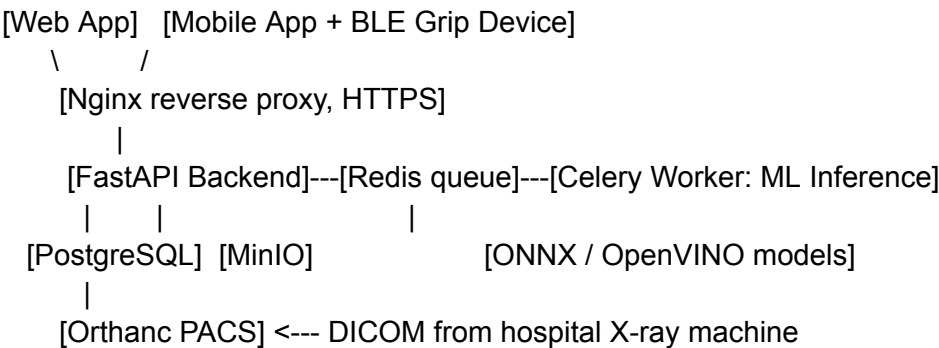
- **Hackathon MVP:** upload, segmentation overlay, ratios, grip entry (simulated or BLE), fusion score, PDF report, history graph.
- **Phase 2:** Orthanc/FHIR integration, KL grade, multi-language.
- **Phase 3:** Prospective OPD pilot, external validation, ABDM integration.

TECHNICAL REQUIREMENTS DOCUMENT (TRD)

1. Architecture Summary

Three layers running in Docker Compose on a single on-prem machine:

- 1. **Clients:** Next.js web app, React Native (Expo) mobile app.
- 2. **Backend:** FastAPI REST API, background workers, ML inference service.
- 3. **Data and infra:** PostgreSQL, Redis, MinIO (image storage), Orthanc (PACS).



2. Tech Stack

Layer	Technology	Reason
Web frontend	Next.js (React), TypeScript, Tailwind CSS, Recharts	Fast, SEO-free dashboard, good charts
Mobile app	React Native with Expo, TypeScript, react-native-ble-plx	One codebase for Android and iOS, BLE support
Backend	Python 3.11, FastAPI, Pydantic, SQLAlchemy, Alembic	Same language as ML code
Task queue	Celery with Redis	Async inference and report generation
Database	PostgreSQL 15	Relational, reliable, encryption options
Object storage	MinIO (S3 compatible)	On-prem image storage
PACS	Orthanc (DICOM C-STORE, DICOMweb)	Lightweight, open source

ML training	PyTorch, MONAI, nnU-Net or U-Net, scikit-learn, XGBoost/LightGBM	Medical imaging standard tools
ML inference	ONNX Runtime, OpenVINO (CPU)	Fast on CPU, no GPU needed
Image processing	OpenCV, pydicom, scikit-image, NumPy	DICOM read, preprocessing, texture features
Explainability	Grad-CAM (pytorch-grad-cam)	Heatmap overlays
Reports	WeasyPrint or ReportLab for PDF; fhir.resources for FHIR	Standardized output
Auth	JWT (access + refresh), bcrypt/argon2, RBAC	Standard, stateless
Infra	Docker, Docker Compose, Nginx	Easy on-prem install
Monitoring	Prometheus + Grafana (optional), structured logs	Basic observability
Hardware	ESP32, HX711 amplifier, load cell, hand dynamometer frame	About INR 1.5k BOM
Testing	Pytest, Jest, Playwright, Postman	Backend, frontend, E2E
CI/CD	GitHub Actions	Automated tests and builds

3. ML Pipeline

1. **Ingest:** receive DICOM/JPG/PNG, normalize intensity, resize.
2. **Quality check:** small CNN classifier (knee AP, acceptable, edges visible).
3. **Segmentation:** U-Net/nnU-Net, 3 classes (0 background, 1 bone, 2 soft tissue).
4. **Muscle feature extraction (rule-based):**
 - Find joint line (row with fewest bone pixels in middle of image).
 - Measure tibial plateau width as the "ruler".
 - Count bone and soft-tissue pixels at thigh and calf levels (0.6 x plateau width above and below joint).
 - Outputs: thigh_soft_to_bone, calf_soft_to_bone, soft_to_plateau, soft_area_ratio.
5. **Bone analysis:** crop proximal tibia ROI, run CNN classifier for osteoporosis risk, optionally add texture features.
6. **Optional:** KL grade classifier.
7. **Fusion:** tabular model (XGBoost or logistic regression) on image features, grip, age, sex, BMI, SARC-F, chair stand, calf circumference. Calibrate with isotonic or Platt scaling.

8. **Staging:** map to AWGS-style stage: No / Possible / Probable / Severe.
9. **Explain:** Grad-CAM overlay and feature table.
10. **Export:** models saved as ONNX, run through OpenVINO.

4. Training Data and Validation

- **Data:** OAI knee radiographs; thigh MRI subset for weak muscle labels; partner-hospital X-rays with paired DXA for fine-tuning.
- **Splits:** patient-level split, no leakage between train and test.
- **Metrics:** AUC (fusion vs grip alone, DeLong test), net reclassification improvement, calibration curve, sensitivity at screening threshold, external site validation.
- **Bias checks:** performance by age group, sex, BMI band.

5. API Design (REST, prefix **/api/v1**)

Auth

Method	Endpoint	Purpose
POST	/auth/login	Email + password, returns access and refresh tokens
POST	/auth/refresh	New access token
POST	/auth/logout	Revoke refresh token
GET	/auth/me	Current user profile

Users (admin)

Method	Endpoint	Purpose
POST	/users	Create user
GET	/users	List users
PATCH	/users/{id}	Update role or status
DELETE	/users/{id}	Deactivate user

Patients

Method	Endpoint	Purpose
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POST	/patients	Register patient
GET	/patients	Search and list
GET	/patients/{id}	Patient details
PATCH	/patients/{id}	Update
GET	/patients/{id}/history	All visits with trend data

Visits and Screening

Method	Endpoint	Purpose
POST	/patients/{id}/visits	Start a new screening visit
GET	/visits/{id}	Visit summary
POST	/visits/{id}/clinical-inputs	SARC-F, chair stand, calf circumference
POST	/visits/{id}/grip	Submit grip trials (BLE or manual)
POST	/visits/{id}/xray	Upload X-ray (multipart)
POST	/visits/{id}/analyze	Trigger AI analysis (async)
GET	/visits/{id}/status	Poll analysis status
GET	/visits/{id}/result	Risk output and features
POST	/visits/{id}/review	Doctor override and notes

Reports

Method	Endpoint	Purpose
POST	/visits/{id}/report	Generate PDF
GET	/reports/{id}/download	Download PDF
GET	/reports/{id}/fhir	FHIR DiagnosticReport JSON

Integrations and Admin

Method	Endpoint	Purpose
POST	/integrations/orthanc/webhook	New study arrived in Orthanc

GET	/devices	Registered grip devices
POST	/devices	Register device
GET	/audit-logs	Audit trail (admin)
GET	/stats/overview	Dashboard statistics

6. BLE Grip Device Spec

- Hardware: load cell, HX711 ADC, ESP32.
- BLE GATT service with a custom characteristic for live weight in kg (float, 10 Hz notifications) and a control characteristic for tare.
- App flow: scan, connect, tare, start trial, capture peak value, repeat 3 times per hand, send best value to backend.
- Calibrate with known weights; store calibration factor on the device.
- Fallback: manual entry and a simulated reading mode for demos.

7. Security and Compliance

- HTTPS (TLS) everywhere, even on local network.
- Passwords hashed with argon2 or bcrypt.
- JWT access token (15 minutes), refresh token (7 days, stored hashed, rotated).
- Role-based access control on every endpoint.
- Encrypted PostgreSQL volume and encrypted MinIO bucket.
- DICOM anonymization for any data used in training.
- Audit log for all reads and writes of patient data.
- DPDP Act 2023: consent record, purpose limitation, data deletion on request, minimal data collection.
- Rate limiting and input validation on all APIs.

8. Hardware and Deployment Requirements

Item	Minimum spec
Server	Intel i5, 16 GB RAM, 256 GB SSD, no GPU
OS	Ubuntu 22.04 LTS
Network	Local LAN; internet optional
Clients	Any modern browser; Android 10+ phone or tablet
Software	Docker and Docker Compose

9. Performance Targets

- Segmentation plus features plus fusion: under 5 s per study.
- API response (non-ML): under 300 ms.
- Report PDF: under 10 s.

APP FLOW (USER JOURNEY AND NAVIGATION MAP)

1. Roles

Technician, Doctor, Admin, Patient (mobile, read-only).

2. Main User Journey (Technician plus Doctor)

Step 1: Login

User opens web or mobile app, enters email and password, and lands on the dashboard for their role.

Step 2: Find or register patient

- Search by name, phone, ID or ABHA.
- If not found, click "New Patient" and fill in name, age, sex, height, weight, contact, and consent checkbox.

Step 3: Start new screening visit

Open patient profile and tap "New Screening".

Step 4: Clinical inputs (optional but recommended)

Enter SARC-F score, 5-chair-stand time, calf circumference. Skippable.

Step 5: Handgrip test

- Mobile: connect to BLE dynamometer, tare, patient squeezes. Repeat 3 trials per hand. App shows live reading and best value.
- Fallback: manual entry.
- Result is compared instantly with the AWGS cutoff (28 kg male, 18 kg female) and shown as Low or Normal.

Step 6: Upload knee X-ray

- Option A: upload file from device.
- Option B: pick study that arrived automatically from Orthanc.
- Quality check runs. If it fails (wrong view, cropped edges), show the reason and ask for a re-upload.

Step 7: AI analysis

Tap "Analyze". Progress screen shows: Segmentation, Measurement, Risk calculation. Takes about 5 seconds.

Step 8: Results screen

- Sarcopenia stage card (No / Possible / Probable / Severe) with probability.
- Osteoporosis risk card (Low / Moderate / High).
- X-ray with segmentation overlay and Grad-CAM toggle.
- Table of measured values against cutoffs.
- Suggested actions (nutrition, physiotherapy, DEXA referral).

Step 9: Doctor review

Doctor can accept, override stage, and add notes. Both AI and doctor values are stored.

Step 10: Report

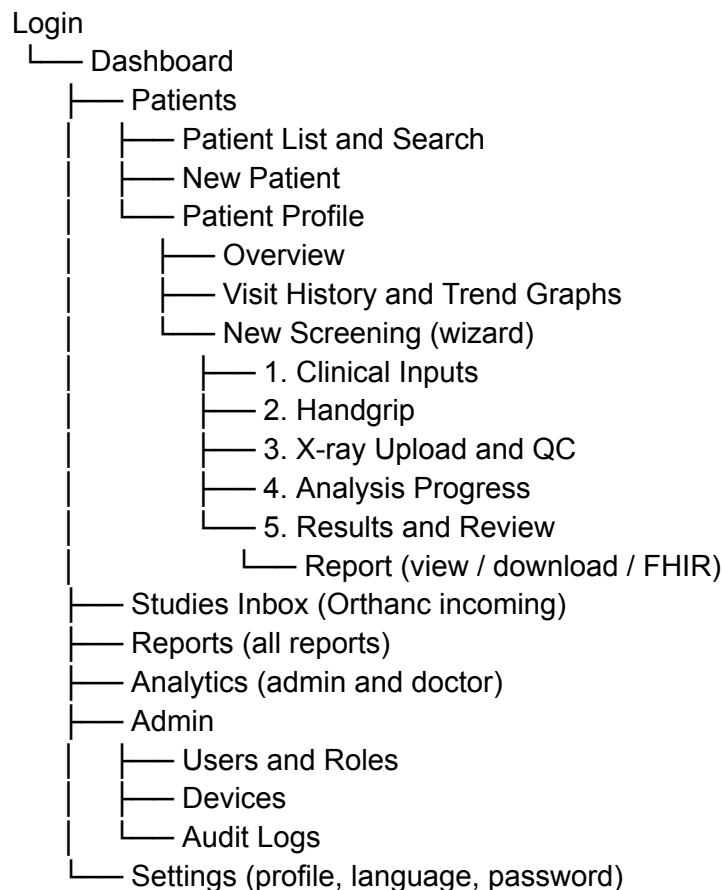
Generate PDF. Download, print, or share. FHIR export is available on web.

Step 11: Follow-up

On a repeat visit, the history graph shows grip, risk and ratio trends over time.

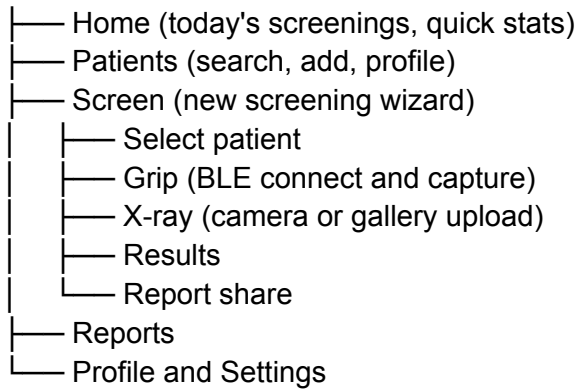
3. Navigation Map

Web App



Mobile App





4. Screen Inventory

1. Login
2. Forgot or reset password
3. Dashboard (role specific)
4. Patient list and search
5. Patient registration form
6. Patient profile
7. Screening wizard (5 steps)
8. BLE device pairing screen
9. Results screen
10. Report preview
11. Trend or history screen
12. Studies inbox
13. Admin: users, devices, audit logs
14. Settings

5. Key Error and Edge Flows

- BLE disconnects: show a reconnect prompt, keep captured trials.
- X-ray fails QC: show the reason and a sample of a good image.
- Analysis fails: retry button, error logged.
- No internet: mobile saves data locally and syncs later.
- Session expired: silent token refresh, otherwise redirect to login.

BACKEND SCHEMA

1. Database: PostgreSQL

Tables

hospitals

Column	Type	Notes
id	UUID, PK	
name	VARCHAR	
city, state	VARCHAR	
created_at	TIMESTAMP	

users

Column	Type	Notes
id	UUID, PK	
hospital_id	UUID, FK to hospitals	
name	VARCHAR	
email	VARCHAR, UNIQUE	
password_hash	VARCHAR	argon2 or bcrypt
role	ENUM(admin, doctor, technician, patient)	
is_active	BOOLEAN	
created_at, last_login_at	TIMESTAMP	

refresh_tokens

Column	Type	Notes
id	UUID, PK	
user_id	UUID, FK to users	
token_hash	VARCHAR	
expires_at	TIMESTAMP	
revoked	BOOLEAN	

patients

Column	Type	Notes
id	UUID, PK	
hospital_id	UUID, FK	
mrn	VARCHAR	hospital record number
abha_id	VARCHAR, nullable	
name	VARCHAR	encrypted field
dob / age	DATE / INT	
sex	ENUM(male, female, other)	
height_cm, weight_kg	FLOAT	
phone	VARCHAR	encrypted
consent_given	BOOLEAN	
consent_at	TIMESTAMP	
user_id	UUID, FK, nullable	links to patient login
created_by	UUID, FK to users	
created_at	TIMESTAMP	

visits

Column	Type	Notes
id	UUID, PK	
patient_id	UUID, FK	
performed_by	UUID, FK to users	
visit_date	TIMESTAMP	
status	ENUM(draft, in_progress, analyzed, reviewed, reported)	
bmi	FLOAT	computed
notes	TEXT	

clinical_inputs

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK, UNIQUE	
sarcf_score	INT	0 to 10
chair_stand_5_sec	FLOAT	
calf_circumference_cm	FLOAT	

grip_measurements

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK	
device_id	UUID, FK, nullable	
hand	ENUM(left, right)	

trial_no	INT	1 to 3
value_kg	FLOAT	
is_best	BOOLEAN	
source	ENUM(ble, manual, simulated)	
captured_at	TIMESTAMP	

devices

Column	Type	Notes
id	UUID, PK	
hospital_id	UUID, FK	
name, ble_address	VARCHAR	
calibration_factor	FLOAT	
last_calibrated_at	TIMESTAMP	

xray_studies

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK	
source	ENUM(upload, orthanc)	
orthanc_study_id	VARCHAR, nullable	
storage_key	VARCHAR	MinIO object path
laterality	ENUM(left, right, unknown)	
qc_passed	BOOLEAN	
qc_reason	VARCHAR	

uploaded_at	TIMESTAMP	
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analysis_results

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK	
xray_id	UUID, FK	
model_version_id	UUID, FK	
mask_storage_key	VARCHAR	segmentation mask
overlay_storage_key	VARCHAR	overlay image
gradcam_storage_key	VARCHAR	
thigh_soft_to_bone	FLOAT	
calf_soft_to_bone	FLOAT	
soft_to_plateau	FLOAT	
soft_area_ratio	FLOAT	
cortical_index	FLOAT, nullable	
kl_grade	INT, nullable	
sarcopenia_prob	FLOAT	
sarcopenia_stage	ENUM(none, possible, probable, severe)	
osteoporosis_prob	FLOAT	
osteoporosis_tier	ENUM(low, moderate, high)	
inference_ms	INT	
created_at	TIMESTAMP	

doctor_reviews

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK	
doctor_id	UUID, FK to users	
final_stage	ENUM	override value
agrees_with_a i	BOOLEAN	
notes	TEXT	
reviewed_at	TIMESTAMP	

reports

Column	Type	Notes
id	UUID, PK	
visit_id	UUID, FK	
pdf_storage_key	VARCHAR	
fhir_json	JSONB	
generated_by	UUID, FK	
generated_at	TIMESTAMP	

model_versions

Column	Type	Notes
id	UUID, PK	
name	VARCHAR	e.g. seg_unet, fusion_xgb

version	VARCHAR	
metrics_json	JSONB	AUC, calibration
onnx_path	VARCHAR	
is_active	BOOLEAN	

audit_logs

Column	Type	Notes
id	BIGSERIAL, PK	
user_id	UUID, FK	
action	VARCHAR	VIEW, CREATE, UPDATE, DOWNLOAD, LOGIN
entity_type, entity_id	VARCHAR / UUID	
ip_address	VARCHAR	
created_at	TIMESTAMP	

2. Relationships

- hospitals 1 to many users, patients, devices
- users 1 to many refresh_tokens, visits (performed_by), doctor_reviews
- patients 1 to many visits
- visits 1 to 1 clinical_inputs
- visits 1 to many grip_measurements (up to 6)
- visits 1 to many xray_studies
- xray_studies 1 to many analysis_results (reruns with new models)
- visits 1 to many doctor_reviews
- visits 1 to many reports
- model_versions 1 to many analysis_results

3. Indexes

- patients(hospital_id, mrn), patients(phone)
- visits(patient_id, visit_date DESC)
- grip_measurements(visit_id)
- audit_logs(user_id, created_at)

4. Authentication Flow

1. User sends email and password to `POST /auth/login`.
2. Backend verifies the hash, creates a 15-minute JWT access token (claims: `user_id`, `role`, `hospital_id`) and a 7-day refresh token. The refresh token hash is stored in `refresh_tokens`.
3. Client stores the access token in memory and the refresh token in secure storage (httpOnly cookie on web, Keychain/Keystore on mobile).
4. Each request carries `Authorization: Bearer <token>`.
5. Middleware validates the token and checks role permissions.
6. On expiry, the client calls `/auth/refresh`. The backend rotates the refresh token (old one revoked).
7. Logout revokes the refresh token.
8. Every login and data access is written to `audit_logs`.

Role Permissions

Action	Technician	Doctor	Admin	Patient
Register patient	Yes	Yes	Yes	No
Run screening	Yes	Yes	No	No
View results	Yes	Yes	Yes	Own only
Override or review	No	Yes	No	No
Generate report	Yes	Yes	No	No
Manage users, devices	No	No	Yes	No
View audit logs	No	No	Yes	No

5. Data Flow (Analysis)

1. Client uploads X-ray, API stores the file in MinIO and a row in `xray_studies`.
2. API runs quick QC. If it passes, it queues a Celery job.
3. Worker loads the image, runs segmentation, feature extraction, bone model, fusion model.
4. Worker saves masks and overlays to MinIO and the result row to `analysis_results`.
5. Client polls `/visits/{id}/status` and gets the result when done.
6. Report generation pulls results, renders PDF and FHIR, saves to `reports`.

IMPLEMENTATION PLAN

Phase 0: Team and Planning (Day 1)

- Split roles: ML, backend, web, mobile and hardware, documentation and pitch.
- Create a GitHub repo with folders: `backend/`, `ml/`, `web/`, `mobile/`, `firmware/`, `docs/`, `infra/`.
- Set up a task board and shared environment variable template.

Phase 1: Project Setup

Backend

- Create a virtual environment, install FastAPI, Uvicorn, SQLAlchemy, Alembic, Pydantic, Celery, Redis client, pydicom, OpenCV, onnxruntime, WeasyPrint.
- Structure: `app/api`, `app/models`, `app/schemas`, `app/services`, `app/workers`, `app/core`.

Web

- `npx create-next-app` with TypeScript and Tailwind. Add axios, react-query, Recharts, react-hook-form, zod.

Mobile

- `npx create-expo-app`, add react-navigation, react-native-ble-plx, expo-image-picker, expo-secure-store, react-query. (BLE needs a dev build, not Expo Go.)

Infra

- `docker-compose.yml` with services: postgres, redis, minio, orthanc, backend, worker, web, nginx.

Phase 2: Authentication

1. Create `users`, `refresh_tokens`, `hospitals` tables with Alembic.
2. Build login, refresh, logout, me endpoints.
3. Add a role-checking dependency in FastAPI.
4. Build login screens on web and mobile with secure token storage and auto-refresh interceptor.
5. Seed an admin, a doctor and a technician account.

Phase 3: Database and Schemas

1. Write all models from Document 4.
2. Generate migrations and apply.
3. Write Pydantic request and response schemas.
4. Add a seed script with a few dummy patients and visits for demos.

Phase 4: Core UI and Layout

Web

- App shell: sidebar, top bar, role-aware menu.
- Pages: Dashboard, Patient List, Patient Profile, Screening Wizard, Results, Reports, Admin.
- Reusable components: risk badge, stat card, data table, stepper, file dropzone.

Mobile

- Bottom tab navigator, patient list, screening wizard screens, results screen.
- Large touch targets and clear typography for clinic use.

Phase 5: Core Features (build in this order)

5.1 Patient and visit management

CRUD endpoints and UI, BMI auto-calculation.

5.2 Grip module

- Firmware: ESP32 reads HX711, applies calibration, sends kg over BLE.
- Mobile: scan, connect, tare, capture 3 trials per hand, mark best.
- Build a simulated grip mode first so the demo never depends on hardware.
- Backend: store trials, compare to AWGS cutoffs.

5.3 X-ray upload and QC

- Upload endpoint, store in MinIO.
- QC: start with simple rules (image size, aspect ratio, edge-pixel check for cropping), then add a small classifier.

5.4 Segmentation and measurement

- Train U-Net on a small labeled set (or use pseudo-labels), export to ONNX.
- Implement the pixel-counting function (joint line, plateau ruler, thigh and calf levels, ratios).
- Save overlay image.

5.5 Osteoporosis module

- Crop proximal tibia, train a CNN classifier, output a calibrated probability.
- If time is short, use texture features with a simple classifier and label it as a prototype.

5.6 Fusion model

- Train XGBoost on tabular features (ratios, grip, age, sex, BMI, SARC-F, chair stand).
- Calibrate, map to staging.
- Compare AUC against grip-only baseline and keep that plot for the pitch.

5.7 Explainability

Grad-CAM overlay and a feature table with cutoffs.

5.8 Reports

- HTML template to PDF with patient info, risk cards, overlay, values vs cutoffs, suggested actions, disclaimer.
- FHIR DiagnosticReport JSON.

5.9 History and trends

Line charts for grip, muscle ratio, sarcopenia probability across visits.

5.10 Doctor review

Override form and storage.

Phase 6: Integrations

- **Orthanc:** run in Docker, configure a Lua script or webhook to call [/integrations/orthanc/webhook](#) on new studies. Demo using a DICOM sender tool (storeescu or Orthanc Explorer upload).
- **FHIR:** build the DiagnosticReport using the fhir.resources library.
- **ABHA (optional):** store the ID field, show as a future-ready hook.
- **SMS/WhatsApp (optional):** skip for MVP.

Phase 7: Testing

Type	Tool	What to test
Unit (backend)	Pytest	Measurement function, AWGS cutoffs, staging logic
API	Pytest and Postman	Auth, permissions, each endpoint
ML	Notebook and scripts	AUC, calibration, grip-only vs fusion, per-sex and per-age results

Frontend	Jest and React Testing Library	Forms, components
E2E	Playwright	Login to report flow
Hardware	Manual	Calibrate with known weights, BLE reconnect
Performance	Locust or timing script	Under 5 s inference on CPU
Security	Manual checklist	Role access, token expiry, no PHI in logs

Phase 8: Deployment

1. Convert models to ONNX and benchmark with OpenVINO on a CPU-only machine.
2. Build Docker images, push to a registry or build on target.
3. On the server: install Docker, copy `docker-compose.yml` and `.env`, run `docker compose up -d`.
4. Set up Nginx with a self-signed or local CA certificate.
5. Enable nightly PostgreSQL and MinIO backups.
6. Build the Android APK (EAS build) and share with the team.
7. Write a one-page install guide and a user manual.

Phase 9: Hackathon Submission Checklist (team leader only)

- Solution write-up built from PRD and TRD
- Architecture diagram
- Demo video: upload X-ray, grip, result, report, trend
- Pitch deck
- GitHub repo link and README
- Slide showing grip-only vs fusion AUC, with honest limits stated
- Clear note on future work: prospective pilot, external validation, regulatory path

Suggested Timeline (7 days example)

Day	Focus
1	Setup, auth, schema
2	Patient and visit CRUD, UI shell
3	Grip module, X-ray upload
4	Segmentation, measurement function

5	Fusion model, osteoporosis module
6	Reports, trends, Orthanc demo
7	Testing, deployment, video, deck